

TFT LCD MODULE
2.7inch 262K 240RGB*432DOTS

MODULE NUMBER: CMS2435TRKSZ2

REVISION: A00

客户确认:

--

开发部	品质部	审核



Document Revision History

Version	Date	Page	Description	Changed By
A00	15th-June-2011	-	First issue	Alex Lee



深圳川马电子有限公司

ShenZhen ChuanMa Electronics Company Limited

Index

Contents	Page
1. LCM Specification	4
2. Mechanical Specification.....	5
3. Pin Descriptions.....	6
4. Electrical Units.....	7
5. AC Characteristics.....	9
6. Optical Specifications.....	10
7. Reliability Test Items.....	14
8. Package(TBD).....	15
9. Handling Precautions.....	15
10. QC.....	16



1. LCM Specification

1.1 Description

CMS2435TRKSZ2 is a transmissive type color active matrix liquid crystal display(LCD) which uses amorphous thin film transistor(TFT) as switching devices. This product is composed of a TFT LCD panel, a drive IC, a FPC, a touch panel, and a LED-backlight unit. The active display area is 2.7 inches diagonally measured and the native resolution is 240*RGB*432. Features of this product are listed in the following table.

1.2 Functions & Features

Table 1.1 Module Functions & Features

Parameter	Value	Unit
LCD Mode	TFT/Transmissive	-
Color Depth	262K	-
Display Resolution	240RGB*432	pixels
Module Size	70.5(V)*41.0(H)*1.95(T)(Exclude FPC)	mm
Active Area (A.A)	59.7(V)*33.9(H)	mm
Pixel Arrangement	RGB-stripe	-
Viewing Direction	6 o' clock	
Display Mode	Normally white	
LCD Controller/Driver	S6D0170	-
IC Package Type	COG	-
MPU Interface	Standard 8080 system 8bit or 16bit parallel interface	-
Power Supply Voltage	2.5~3.3	V
Back-light	White LED*4	PCS





3. Pin Descriptions

Pin No.	Symbol	I/O	Functional	Remark
1~2	GND	Power	System power ground.	
3	FLM	O	Output pin for test	
4	CSB	I	Chip select pin	
5~6	VCI	Power	Power supply pin, VCI=2.4~3.3(V)	
7	RS	I	Register select signal. Low: Index register or internal status is selected. High: Control register is selected..	
8~12	D0~D4	I/O	Data bus	
13	GND	Power	System power ground.	
14~18	D5~D9	I/O	Data bus	
19	GND	Power	System power ground..	
20~25	D10~D15	I/O	Data bus	
26~27	VDD3	Power	Logic interface power supply VDD3=1.65~3.3(V)	
28	WR	I	80-system : WR (write strobe signal)	
29	RD	I	80-system : RD (write strobe signal)	
30	GND	Power	System power ground.	
31	RESET	I	Reset signal pin	
32	GND	Power	System power ground.	
33	LEDA	Power	LED Power supply	
34~37	LEDK1~K4	Power	LED Power supply	
38~39	GND	Null	System power ground.	

Note: 80-16BIT data bus use DB[15:0] ;80-8BIT data bus use DB[15:8].Interface is selected by initial code.



4. Electrical Units

4.1 Absolute Maximum Ratings

The absolute maximum ratings are list on Table 4.1. When used out of the absolute maximum ratings, the LCM may be permanently damaged. Using the LCM within the following electrical characteristics limit is strongly recommended for normal operation. If these electrical characteristic conditions are exceeded during normal operation, the LCM will malfunction and cause poor reliability.

Table 4.1 Module Absolute Maximum Ratings

Item	Symbol	Unit	Value	Note
Power Supply Voltage (1)	Vdd	V	-0.3 to +4.2	
Power Supply Voltage (2)	VGH ~ VSS	V	-0.3 to +18.0	
Power Supply Voltage (3)	VSS ~ VGL	V	-0.3 to +18.0	
Operating Temperature	Top	°C	-20 to +60	
Storage Temperature	Tst	°C	-30 to +70	
Operating Humidity	Hop	%(RH)	10~90	

4.2 Electrical characteristics (Ta=25°C)

Table 4.2:DC Characteristic (Vcc = 2.3 ~ 3.3V)

Item		Symbol	Condition	Min.	Type.	Max.	Unit
Supply Voltage	Logic	Vdd	---	2.5	3.0	3.3	V
Input Voltage	H level	V _{IH}	---	0.8V _{dd}	---	V _{dd}	V
	L level	V _{IL}		0	---	0.2V _{cc}	
Current Consumption		I _{DD}	With internal voltage generation; VDD=3.0V; Tamb=25°C;	---	---	TBD	mA
LCD Driving Voltage		VOP	---	---	TBD	---	V



4.3 Back-light Specification

Table 4.3 Back-light Characteristics

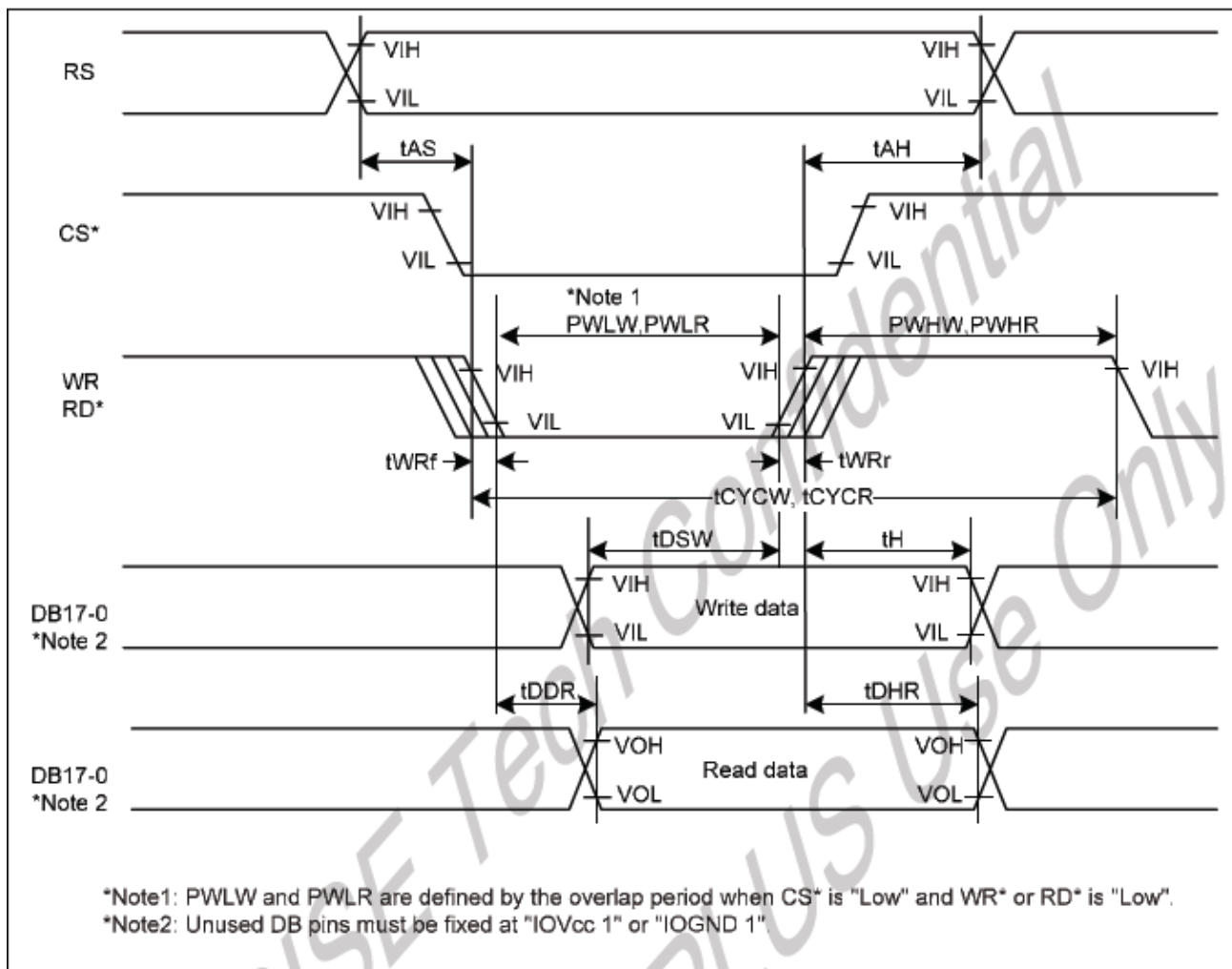
Item	Symbol	Conditions	Min.	Type.	Max.	Unit
Supply Voltage	VF	Only Backlight	3.0	3.2	3.5	V
Supply Current	IF		60			mA
Average Brightness	IV	Backlight Current IF=60mA	3000	3500	-	Cd/m2
CIE Color Coordinate (Without LCD)	X	Backlight Current IF=60mA		0.26		—
	Y			0.30		
Uniformity	B	Backlight Current IF=60mA	80	—	—	%
Color	White					

Note: 4LEDs in parallel connection.



5. AC Characteristics

5.1 80-system bus interface operation





6. Optical Specifications

Optical characteristics are determined after the unit has been 'ON' and stable for approximately 30 minutes in a dark environment at 25°C. The values specified are at an approximate distance 50cm from the TFT-LCD surface at a viewing angle of Φ and θ equal to 0°.

Measurement condition: Refer to next pages (C-light source, Halogen Lamp)

*1): with Polarizer *2): without Polarizer *3): Only Color Filter glass

Item		Symbol	Conditions	Specifications		
				Min.	Typ.	Max.
Transmittance		T%	Viewing normal angle $\theta_x = \theta_y = 0^\circ$	6.7		
Contrast Ratio		CR		150	250	-
Response Time		T_R		-	15	30
		T_F		-	35	50
Chromaticity	Red	X_R		0.606	0.636	0.666
		Y_R		0.298	0.328	0.358
	Green	X_G		0.272	0.302	0.332
		Y_G		0.548	0.578	0.608
	Blue	X_B		0.102	0.132	0.162
		Y_B		0.107	0.137	0.167
	White	X_W		0.274	0.304	0.334
		Y_W		0.317	0.347	0.377
Viewing Angle	Hor.	θ_{x+}	Center CR ≥ 10	-	45	-
		θ_{x-}		-	45	-
	Ver.	θ_{y+}		-	35	-
		θ_{y-}		-	15	-



深圳川马电子有限公司

ShenZhen ChuanMa Electronics Company Limited

Notes : 1. Contrast Ratio(CR) is defined mathematically as :

$$\text{Contrast Ratio} = \frac{\text{Surface Luminance with all white pixels}}{\text{Surface Luminance with all black pixels}}$$

2. Surface luminance is the center point across the TFT-LCD surface 500mm from the surface with all pixels displaying white. For more information see FIG 1.
3. Response time is the time required for the display to transition from white to black(Rise Time, Tr) and from black to white(Falling Time, Tf). For additional information see FIG 3.
4. Viewing angle is the angle at which the contrast ratio is greater than 10. The angles are determined for the horizontal or x axis and the vertical or y axis with respect to the z axis which is normal to the TFT-LCD surface. For more information see FIG 4.
5. Optimum contrast is obtained by adjusting the TFT-LCD Threshold voltage(Vth & Vsat)

FIG. 1 Optical Characteristic Measurement Equipment and Method

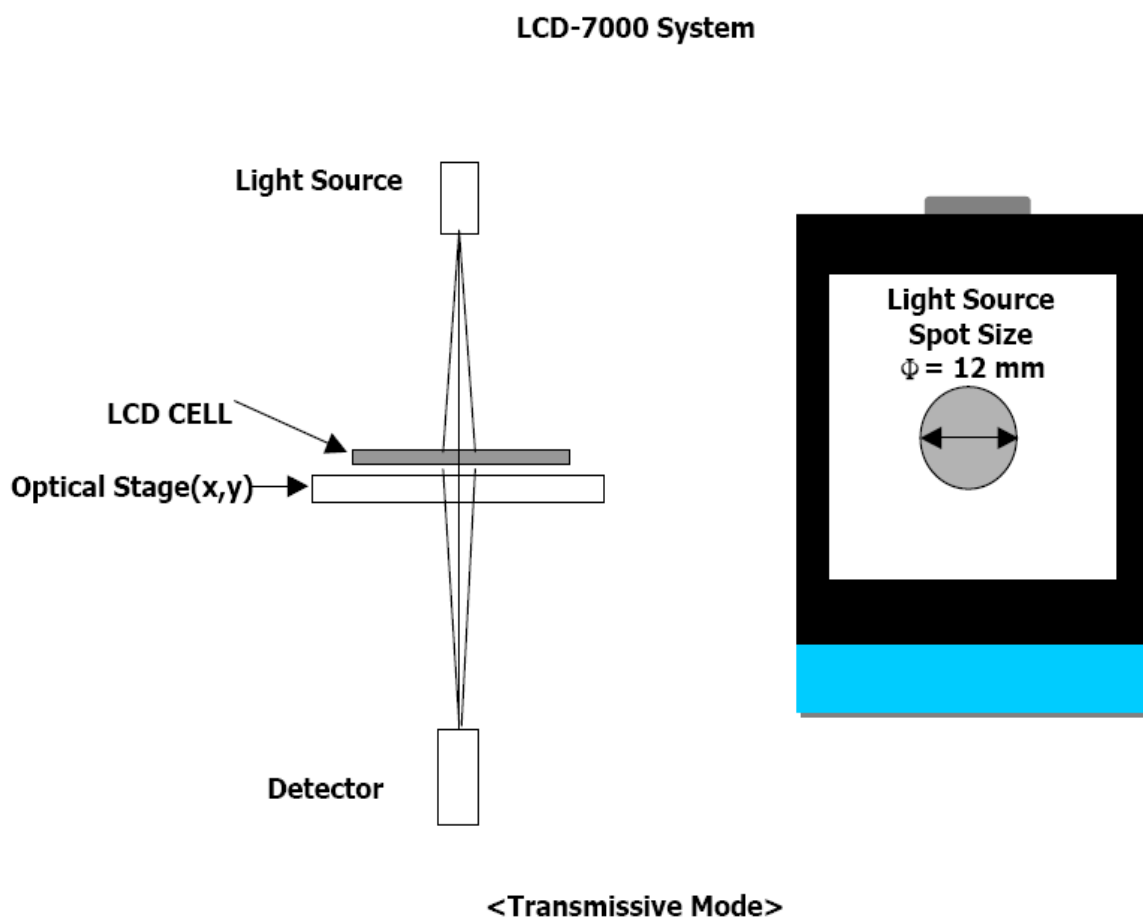




FIG. 2 The definition of V_{th} and V_{sat}

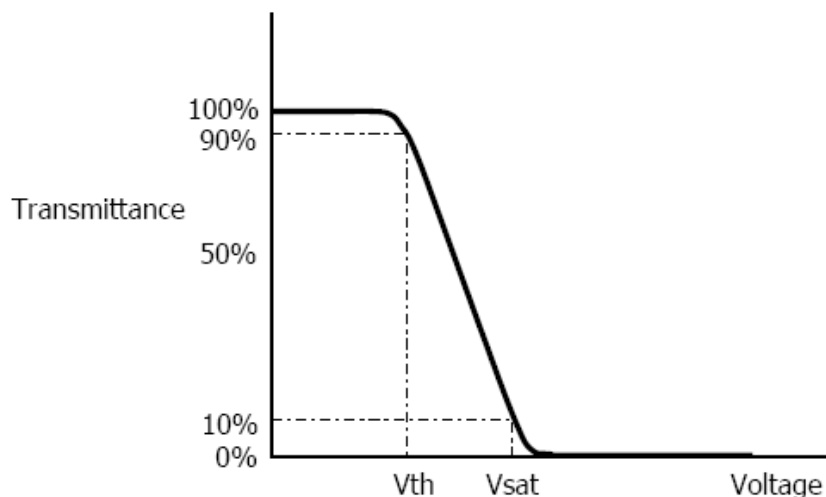
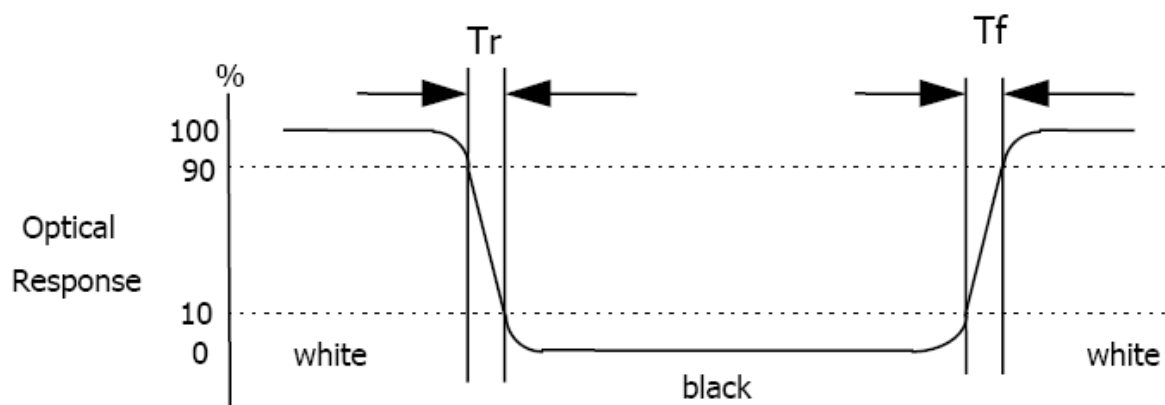


FIG. 3 The definition of Response Time

The response time is defined as the following figure and shall be measured by switching the input signal for "black" and "white".



* Voltage conditions for Response time

V_{gate} : 19V DC

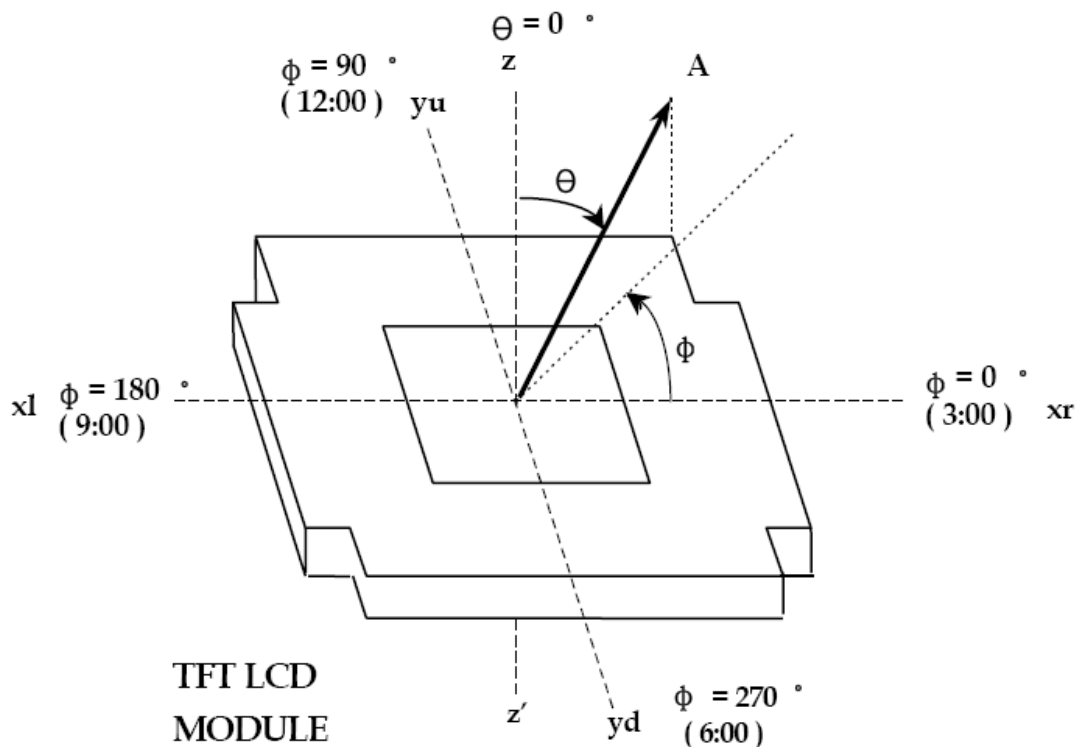
V_{data} : 0V~3.3V DC

V_{com} : 0V (Ground)



FIG. 4 The definition of viewing angle

<dimension of viewing angle range>



7. Reliability Test Items

No.	Test Item	Test Condition	Check Time
1	High temp storage	T= 80℃	240 hrs
2	Low temp storage	T= -30℃	240 hrs
3	High temp operation	T= 70℃	240 hrs
4	Low temp operation	T= -20℃	240 hrs
5	High temp & high humidity	T=60℃ H=90%	240 hrs

Reliability Test Criteria:

Display function should be no change under normal operating condition.



8. Package(TBD)

9. Handling Precautions

9.1 Safety

The liquid crystal in the LCD is poisonous. Keep away from your mouth and eyes. If the liquid crystal contacts with your skin, mouse or clothes, use soap to wash it off immediately.

9.2 Handling

- i. The LCD panel is made by thin glass. Prevent the panel from mechanical shock or putting excessive force on its surface.
- ii. The polarizer attached on the display is very easy to be damaged, handle it with special attention.
- iii. To avoid contamination on the display surface, do not touch the display surface with bare hands.
- iv. The transparent electrodes may be disconnected if you use the LCD panel under dew-condensing environment.
- v. The characteristics of the semiconductor devices may be affected when they are exposed to light, possibly resulting in malfunctioning of the ICs. To prevent such malfunctioning of the ICs, make sure the application and the mounting of the panel are designed so that the IC is not exposed to light.

9.3 Static Electricity

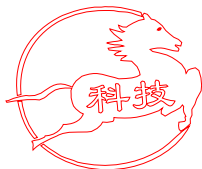
Ground soldering iron tips, tools and testers when you operate. Also ground your body when handling the products and store the products in an anti-electrostatic container.

9.4 Storage

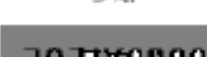
Store the products in a dark place where the temperature is within the range of $25\pm 10^{\circ}\text{C}$ and with low humidity (65%RH or less). Do not store the LCD product in an atmosphere containing organic solvents or corrosive gases.

9.5 Cleaning

Do not wipe the polarizer with dry cloth, as it might cause scratching. Wipe the polarizer with a soft cloth soaked with petroleum IPA. Other chemical might damage the panel.



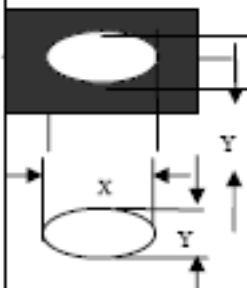
10. QC

主题:		文件编号: CMFI-8202-02			
彩屏模组检验规范		版本/版次: A/0			
		页数/页码: 1/7			
1. 目的: 为了规范彩屏产品的检验,保证彩屏产品品质.					
2. 范围: 适用于所有彩屏产品及 CSTN 产品的检验.(客户有特殊要求的依客户要求)					
3. 职责与权限: 品质: 对此检验标准进行维护及修订,并依此检验标准进行检验. 生产: 依照本检验进行检验.					
4. 抽样方案:					
4.1.					
检验项目		抽样计划			
外观、电讯		GB/T2828. 1-2003 II 正常单次抽样方案			
尺寸		10PCS/批			
4.2 检验者需戴静电环、两手八个手指套,若为无贴附偏光片的 LCD 则十指全戴手指套.					
4.3 检验者以目视检查或与尺寸对照表比对.					
4.4 电性测试使用测试治具、显示画面请依样品.					
4.5 检验者目视与产品距离 30 公分, 检验方向为垂直线前后左右 60° .					
4.6 密集定义:1mm 以内若有两点(圆形物)、两条(线状物)或圆形物与线状物, 则称之为密集.					
5.允收水准:					
主缺: 0.4 次缺:0.65					
异常项目		判定			
外观、电讯		GB/T2828. 1-2003 II 正常单次抽样方案判定			
尺寸		1PCS 不良则批退			
6.判定标准及检验方法:					
项目	缺点类别	不良定义	不良现象	判定标准	检验方法
1.电讯	1-1(主缺)	LCD 点亮后有整行(或列)或字段点阵不显的现象	少划 	NG	用测架点亮LCD后目视
	1-2(主缺)	在显示字体画面时有不该显示的地方显示(画面乱七八糟)	乱讯/显示乱码 	NG	用测架点亮LCD后目视
	1-3(主缺)	非显示区有较模糊的显示	多划 	NG	用测架点亮LCD后目视



深圳川马电子有限公司

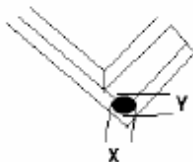
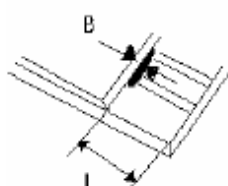
ShenZhen ChuanMa Electronics Company Limited

主题:			文件编号: CMWI-8202-02													
彩屏模组检验规范			版本/版次: A/0													
			页数/页码: 2/7													
6.判定标准及检验方法:																
项目	缺点类别	不良定义	不良现象	判定标准	检验方法											
1.电讯	1-4.(主缺)	比对样品有字深字淡现象	显示深浅	VOP±0.2V OK VLCD±0.2V OK	在测试条件一致的情况下量测产品 Vop 值是否在偏差范围内											
			IC:HX8309 深													
			IC:HX8309 OK													
		IC:HX8309 浅														
	1-5.(主缺)	通电后 LCD 无动作或无画面	无显示	NG	目视											
1-6.(主缺)	点亮后 LCD 显示画面时有时无	显示不稳	NG	点亮后目视												
1-7.(主缺)	在显示过程中有些画面没显示	少画面	NG	通电后测试												
1-8.(主缺)	LCD 视角与样品不一致	LCD 视角错误	NG	用测架点亮后目视												
2.外观	2-1.(主缺)	通电后 LED 不亮	背光不亮	NG	目视											
	2-2.(主缺)	通电后背光的电流值超出规定电流值	背光大电流	NG	目视											
	2-3.(次缺)	产品通电后显示区内不能有亮眼,漏光	亮眼,漏光	NG	目视											
	2-4.(主缺)	通电后背光颜色偏黄或偏红与样品,规格书不符	背光颜色不符	与样品,规格书一致	点亮背光后目视											
	2-5.(次缺)	背光不发光区域进入可视区	露黑边	参照工程图纸判定	点亮产品后目视											
	2-6.(次缺)	通电后背光亮度与发光均匀度达不到规格书要求	亮度不够 发光不均匀	与样品,规格书一致	点亮背光后目视											
	2-7.(主缺)	在 LCD 上有擦拭不掉的点状物	黑点、白点 	<table><tr><td>尺寸 (直径Φ)</td><td>允许个数</td></tr><tr><td>Φ≤0.1mm</td><td>不计</td></tr><tr><td>0.1<Φ≤0.15</td><td>2</td></tr><tr><td>0.15<Φ≤0.20</td><td>1</td></tr><tr><td>Φ>0.20</td><td>0</td></tr><tr><td colspan="2">备注: Φ=(X+Y)/2</td></tr></table>	尺寸 (直径Φ)	允许个数	Φ≤0.1mm	不计	0.1<Φ≤0.15	2	0.15<Φ≤0.20	1	Φ>0.20	0	备注: Φ=(X+Y)/2	
尺寸 (直径Φ)	允许个数															
Φ≤0.1mm	不计															
0.1<Φ≤0.15	2															
0.15<Φ≤0.20	1															
Φ>0.20	0															
备注: Φ=(X+Y)/2																
主题:			文件编号: CMWI-8202-02													
彩屏模组检验规范			版本/版次: A/0													
			页数/页码: 3/7													



深圳川马电子有限公司

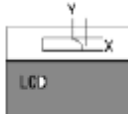
ShenZhen ChuanMa Electronics Company Limited

主题:				文件编号: CMWI-8202-02			
彩屏模组检验规范				版本/版次: A/0			
				页数/页码: 4/7			
6.判定标准及检验方法:							
项目	缺点类别	不良定义	不良现象	判定标准			检验方法
2.外观	2-10.(次缺)	LCD 有崩裂现象	D.崩角	判定标准			目视 目测镜 菲林卡
				X	Y	Z	
				≦2	≦1.5	≦t	
				注:如果崩裂位置与最近的ITO引线距离小于ITO引脚的宽度,则按崩ITO引脚标准			
	2-11.(主缺)	LCD 有裂痕现象	裂痕	任何区域有任何裂痕都不允许			目视
	2-12.(次缺)	LCD 在裂片时有凹凸不平现象	A.切裂不良	判定标准			目视 目测镜 菲林卡
				1.长度不计 2.B≦0.30			
			B.切裂不良	判定标准			
		L,W 超出图纸标注尺寸拒收					
	2-13.(次缺)	框胶有杂质变窄等不良现象	1. 造成框胶变窄及框胶杂质,气泡超过 1/3 框胶宽度 2. 框胶偏移进入显示区	NG			目视
	2-14.(次缺)	保护膜脱离 LCD	1. 保护膜翘起,顶破,刮破进入可视区 2. 保护膜胶落及保护膜难撕开	拒收			目视
	2-15.(次缺)	LCD 整体底色一致,但与样品不同	底色不符	比照限度样本验收			在同一视角下用样品或色卡比对



深圳川马电子有限公司

ShenZhen ChuanMa Electronics Company Limited

主题:				文件编号: CMWI-8202-02			
彩屏模组检验规范				版本/版次: A/0			
				页数/页码: 5/7			
6.判定标准及检验方法:							
项目	缺点类别	不良定义	不良现象	判定标准		检验方法	
2.外观	2-16.(次缺)	贴附偏光片时不能有歪斜	偏光片贴附	依工程图要求公差判定		目视	
	2-17(次缺)	COG IC 有崩角现象	IC 崩角  $\Phi=(X+Y)/2$	1. IC 崩角 $\Phi \cong 0.5\text{mm}$,崩一角 2. 电性 OK 3. 不存在裂纹 以上同时符合允收		显微镜	
	2-18.(次缺)	FPC 贴附元件后不能有不良现象	1.划伤	以 2-8 项刮伤规格判定且不能进入布线区		目视	
			2.外观尺寸	符合工程图要求			
			3.顶伤,刮破	不允许			
			4.金手指氧化,脏,折痕	不允许,必要时依限度样判定			
			5.虚焊假焊少焊锡连锡	不允许,以《LCM 成品检验规范》2.15、2.20 进行判定			
			6.元件焊接	必须正确,不影响组装			
	2-19(次缺)	涂硅胶时不能有气泡,超高于偏光片等现象	涂胶不良	1.涂胶面积必须盖住 ITO 引线 允收 2.涂胶高度不超过偏光片表面 允收 3.涂胶必须均匀,光滑,无气泡,针孔 允收		目视	
	2-20(次缺)	黑色遮光胶纸/黄色胶纸未贴附好	胶纸贴歪等不良	1.胶纸粘性良好,不能有翘起,歪斜,皱折,撕裂等现象 允收 2.颜色,尺寸,贴附位置依工程图纸要求或参照样品进行判定		目视	
	2-21.(次缺)	在触摸屏上有擦试不掉的点状物	触摸屏有刺伤、脏点 圆形物	点直径	判定	备注	目视
				$\Phi \cong 0.1$	不计	两个点	
				$0.10 \cong \Phi \cong 0.15$	$\cong 2$ 个	之间的	
				$0.15 < \Phi \cong 0.2$	1 个	距离要	
				$0.2 < \Phi$	0	求大于 10mm	



深圳川马电子有限公司

ShenZhen ChuanMa Electronics Company Limited

主题:

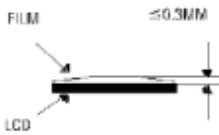
彩屏模组检验规范

文件编号: CMWI-8202-02

版本/版次: A/0

页数/页码: 6/7

6.判定标准及检验方法:

项目	缺点类别	不良定义	不良现象	判定标准			检验方法
2.外观	2-22(次缺)	在触摸屏上有擦试不掉的线状物	刮伤、纤维、线状物	参照 2-8 项标准进行判定			目视
	2-23(主缺)	在触摸屏上有崩及裂痕	崩及裂痕	参照 2-9 A 及 2-11 进行判定			目视
	2-24(主缺)	LCD 有漏液现象	漏液	拒收			目视
	2-25(主缺)	LCD 盒内有气泡	盒内气泡	拒收			目视
	2-26(主缺)	FILM 表面有折痕	FILM 表面折痕	拒收			目视
	2-27(次缺)	触摸屏上有牛顿环及干涉线	牛顿环	≡ 整屏的 1/3 允收 1 个			迎光 目视
			干涉线	≡ 1/4 触摸屏检查面积允收			
	2-28(次缺)	触摸屏上有鼓的现象	触摸屏鼓 	膜鼓起在 0.3MM 以下允收			目镜或塞规
	2-29(次缺)	触摸屏 FILM 表面上有鱼眼气泡	FILM 表面有鱼眼气泡	直径	允收个数	备注	目视 目镜 菲林卡
				Φ ≦ 0.15	不计	两个气泡之间的距离 ≧ 10mm	
				0.15 < Φ ≦ 0.25	允收 ≦ 2 个		
				0.25 ≦ Φ	拒收		
	2-30(主缺)	在组装时造成的不良现象	模组组装与组装说明文件不符	拒收			目视
			组装不符合工艺要求的(固定双面胶纸或遮光胶纸未贴等)	拒收			
			部件间连接不牢(玻璃与背光源脱离等)	拒收			
			LCM 倾斜或偏位(超出允许的规格)	拒收			



深圳川马电子有限公司

ShenZhen ChuanMa Electronics Company Limited

主题: 彩屏模组检验规范				文件编号: CMWI-8202-02 版本/版次: A/0 页数/页码: 7/7	
2. 外观	2-30(主缺)	在组装时造成的不良现象	组装后的 LCM 的 FPC 向后弯折在背光板上点亮的同时从正面看有阴影	拒收	目视
6.判定标准及检验方法:					
项目	缺点类别	不良定义	不良现象	判定标准	检验方法
2.外观	2-31(次缺)	贴附转移纸时的不良现象	粘附符力不够撕起保护膜	拒收	目视
			贴附位置不符合规格要求	拒收	
			贴附边缘 LCD 边缘角度大于 ± 15 度	拒收	
	2-32(次缺)	喷码时的不良现象	不符合文件要求 字体不清晰无法识别	拒收	目视
	2-33(主缺)	LCD 视角错或差	视角错或差	依限度样板式验收	目视
	2-34(次缺)	无	偏光片贴附不良	1. 贴附位置符合工程图纸要求的公差 OK 2. 偏光片必须贴到框胶边框 OK (注:以上条件需同时满足) 3. 偏光片翘起按偏光片气泡判定	目视
	2-35(主缺)	引脚未置于焊盘中心位置	引脚偏移	参照《LCM 成品检验规范》2-22 进行检验	目视
	2-36(主缺)	引角锡盘吃锡高度	着锡量	参照《LCM 成品检验规范》2-23 进行检验	目视
	2-37(主缺)	连接器触点粘锡	触点有锡	参照《LCM 成品检验规范》2-24 进行检验	目视
	2-38(主缺)	连接品 pin 脚剥离主 FPC	连接器剥离	参照《LCM 成品检验规范》2-25 进行检验	手轻掰在显微镜下目视
3.特性	2-39(主缺)	连接器 pin 脚歪斜变形	连接器变形	参照《LCM 成品检验规范》2-26 进行检验	在显微镜下目视
	3.1(主缺)	对位面积未达到标准要求	参照<<LCM 成品检验规范>>3-2		在显微镜下目视/目视
	3.2(主缺)	有效导电粒子有破碎不良	参照<<LCM 成品检验规范>>3-1		在显微镜下目视